

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Computer Science Engineering	Indian Institute of Technology Gandhinagar	2024-Present	8.58
Class XII	Physics, Chemistry, Maths	Pankaj Junior College	2022-24	94.17
Class X		Pankaj Secondary & Higher Secondary School	2020-22	98.00

PROJECTS

- **Single Photon Image Reconstruction**

[Jan'26-May'26]

(Advisor: Shanmuganathan Raman, IIT Gandhinagar) | [Project Link](#)

- Achieved 34.00 dB PSNR and 0.883 SSIM on 3 held-out scenes by engineering a 4-phase pipeline : naive summation -> flat CNN -> UNet -> 237M-parameter ResUNet with Guided Attention Gates - delivering a +20.68 dB gain over the analytical baseline
 - Designed residual conv blocks, attention-gated skip connections, and a hybrid Charbonnier + MS-SSIM + VGG + edge loss with deep supervision annealed 0.4->0 to reconstruct clean RGB images from binary single-photon measurements
 - Trained on 1,850 samples with BF16 mixed precision and cosine warm restarts; each phase benchmarked on consistent held-out scenes, pushing PSNR from 13.32 dB to 34.00 dB
- **StayEase: Hostel Management System**

[Jan'26-May'26]

(Advisor: Yogesh K. Meena, IIT Gandhinagar) | [Project Link](#)

- Designed a normalized relational schema with 18+ tables, cascading foreign keys, CHECK constraints, and a computed view, ensuring referential integrity across room allocation, complaints, and inventory workflows
 - Engineered 25+ RESTful APIs enforcing JWT-based role-based access control for 4 distinct user roles with automatic token expiry and dual audit logging across file and database
 - Prevented concurrent overbooking under race conditions using MySQL row-level locking, validated via Locust stress tests across 31,294 requests with 0 failures
- **UrbanNest: Machine Learning-Based House Rent Prediction System**

[Jan'26-May'26]

(Advisor: Nipun Batra, IIT Gandhinagar) | [Project Link](#)

- Engineered an end-to-end rental price prediction system across 4 Indian cities on 13,900+ listings, achieving Rs 7,900 CV MAE using a Random Forest with 14 features including geo-coordinates and listing metadata
 - Reduced hyperparameter search time by 25% over Grid Search by implementing Bayesian Optimization (Optuna) alongside Grid and Random Search - all benchmarked across 60 trials with 5-fold CV and logged via Trackio
 - Deployed a Streamlit app on Hugging Face Spaces via Docker with auto-resolved locality coordinates from 700+ mapped neighbourhoods, eliminating all manual coordinate input

TECHNICAL SKILLS

- **Programming Languages:** C++, C, Python, HTML, CSS, Javascript (basic)

• **Libraries:** Pandas, Numpy, Matplotlib, Pytorch, Flask

• **Tools:** Git, Github, Linux (Ubuntu)

• **Database:** MySQL